

Post-Foundations Validation Release for HAOS-IIP

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Abstract

This paper freezes release 50.3 of HAOS-IIP as a small post-foundations validation release immediately after 50.2. Its purpose is narrow and explicit. Release 50.2 closed the remaining bounded theorem-ladder cleanup items F2--F4, so the main bottleneck is no longer internal foundations hygiene. Release 50.3 records the resulting phase change. The scalar branch now has a bounded validated continuum-facing contract on a nontrivial tested class, while the vector side has a derived active physical-sector restriction, explicit refinement comparison maps, and measurable branch-identity diagnostics, but still lacks a numerically closed continuum-facing coexact branch. No new numerics are introduced, no disorder-threshold bundle is frozen, and no stronger continuum claim is made. The role of this paper is to state what is now treated as structurally closed, what current numerics already validate, and what exact validation stack still controls the authority boundary.

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1 Scope

Release 50.3 is intentionally small.

It does not add:

- new theorem steps beyond T1--T8;
- new foundational cleanup items beyond F1--F4;
- new scalar or vector result bundles;
- stronger continuum claims.

It does add:

- a frozen post-50.2 validation boundary;
- an explicit statement of what is now treated as structurally closed;
- a compact current validated read for the scalar and vector sides;
- an ordered validation stack for the next real authority tests.

2 What Is Now Structurally Closed

After releases 50.1 and 50.2, the following are treated as closed in the bounded sense used by the repository:

1. the abstract derivation ladder T1--T8;
2. the scalar-locality refinement F1;
3. the incidence-normalization refinement F2;
4. the channel-completeness / no-extra-bridge refinement F3;
5. the Hilbert-complex upgrade F4.

So the main open bottleneck is no longer

what operator should the repository mean?

It is now

does the already-derived active physical branch survive validation strongly enough to support bounded continuum language?

3 Current Validated Read

3.1 Scalar branch

The scalar branch already carries a real bounded validation result. The kernel-Laplacian convergence artifacts now support the following compact read:

- the cubic interior control remains clean;
- weakly perturbed point sets remain convergent in the local-kernel regime;
- reflected Dirichlet and Neumann boundary controls also remain convergent;
- wider kernels remain the weak point and are not the main evidence.

So the scalar branch is no longer only a formal target. It is the strongest currently validated continuum-facing part of the repository.

3.2 Vector / active coexact branch

The vector side is stronger than it was before 50.1, but it is still not numerically closed.

What is now validated is:

- the harmonic / coexact split is explicit rather than inferred;
- the harmonic-to-coexact gap is measured directly;
- the active coexact sector has explicit refinement comparison maps J_n ;
- transported overlap, principal-angle alignment, and scaled-eigen drift now provide measurable branch-identity diagnostics.

At the same time, the present sector-window and transport artifacts still support the following bounded read:

- the raw low L_1 floor remains harmonic;
- the coexact branch is real but sits above that floor by a finite scaled gap;
- defect and mild-disorder backgrounds can stabilize or compress the active branch substantially more than the clean baseline;
- the clean baseline remains weak and mixing-prone on the tested transported low window.

So the present vector conclusion is not that a transverse continuum branch has been established. The bounded conclusion is only that the derived active coexact branch is now measurable as a transported physical object, while its clean-background refinement identity remains open.

4 The Post-Foundations Validation Stack

Release 50.3 records the correct validation order after 50.2.

4.1 V1: active-branch identity under refinement

The first validation task is to determine whether the same low active coexact family survives as the same physical object across refinement.

Success means:

- transported low-window modes retain strong overlap;
- principal-angle alignment remains strong;
- scaled-eigen drift stays bounded without arbitrary relabeling.

Failure means that branch identity continues to reshuffle under refinement even after the physical-sector restriction is fixed.

4.2 V2: symmetry-breaking / disorder-threshold validation

The second task is to determine whether the weakness of the clean baseline is mainly a symmetry-degeneracy problem or a deeper failure of active-branch survival.

Success means there exists a bounded small-disorder regime in which branch identity becomes robust while staying recognizably on the same active coexact sector.

Failure means either that no bounded disorder regime stabilizes the branch, or that apparent stabilization only occurs after such strong perturbation that the original branch identity is no longer credible.

4.3 V3: harmonic recontamination and branch purity

The third task is to test whether the active branch remains physically coexact under refinement rather than drifting back into harmonic or mixed sectors.

Success means the tested low window stays inside the active coexact sector to leading order, with bounded harmonic leakage.

Failure means repeated low-window recontamination by harmonic or mixed content as the refinement grows.

4.4 V4: scaling closure on the active window

The fourth task is to determine whether one coherent scaling family a_n stabilizes the same active window across refinement.

Success means one bounded scaling law keeps the same tracked window finite and comparable.

Failure means the branch only appears stable after repeated window-dependent or case-dependent rescaling.

4.5 V5: universality checks

Only after V1--V4 should the repository ask whether the current scalar and active-sector contracts survive bounded operator-family and substrate-family variation.

4.6 V6: effective-law and response closure

Only after branch identity, threshold stability, purity, scaling closure, and bounded universality are in hand should the repository ask whether the same validated branch closes onto one effective-law family and supports downstream response or geometry-like questions.

5 What Already Counts as Validation

Release 50.3 says that the repository already has enough evidence to state the following honestly:

1. the post-T8 program is no longer blocked by missing foundations cleanup;
2. the scalar branch has bounded operator-level continuum control on a nontrivial tested class;
3. the vector branch now has a real physical-sector restriction and real refinement-comparison machinery;
4. branch identity can now fail in specific measurable ways instead of being discussed only rhetorically;
5. mild disorder and some defect backgrounds already suggest that the weak clean-baseline case is not the same thing as a total failure of the active-sector idea.

Those are meaningful validation gains.

6 What 50.3 Still Refuses To Say

Release 50.3 does not justify any of the following:

- a closed continuum vector claim;
- a Maxwell-like continuum statement;
- universality of the active branch;
- an effective PDE or action law on the vector branch;
- geometry-like or response-like closure on the vector branch.

Those still have to be earned by the validation order above.

7 Conclusion

Release 50.3 marks the phase change from foundations-building to validation.

After 50.2, the clean question is no longer whether HAOS-IIP can derive a bounded harmonic operator ladder. It can.

The clean question is whether the already-derived active physical branch survives refinement, threshold perturbation, purity checks, and scaling closure strongly enough to support bounded continuum language.

That is the whole point of 50.3.